2.12 Inverse Functions

Worked Solutions 1–39

1 a)

One-to-one so f^{-1} exists.

1 b)

Many-to-one so f^{-1} does not exist.

1 c)

One-to-one so f^{-1} exists.

1 d)

Many-to-one so f^{-1} does not exist.

1 e)

Minimum at $(-2, -4)$.

One-to-one so f^{-1} exists.

2 a)

$$\begin{aligned} y &= 1 - 2x \\ x = 1 - 2y \Rightarrow 2y &= 1 - x \Rightarrow y = \frac{1}{2} - \frac{1}{2}x \\ f^{-1}(x) &= \frac{1}{2} - \frac{1}{2}x = \frac{1-x}{2}. \end{aligned}$$

So $f^{-1}(x) = \frac{1-x}{2}$, $x \in \mathbb{R}$.

2 b)

$$\begin{aligned} y &= \frac{x+3}{4} \\ x = \frac{y+3}{4} &\Rightarrow 4x = y + 3 \Rightarrow y = 4x - 3 \\ f^{-1}(x) &= 4x - 3. \end{aligned}$$

So $f^{-1}(x) = 4x - 3$, $x \in \mathbb{R}$.

2 c)

$$\begin{aligned} y &= \frac{7}{2x} \\ x = \frac{7}{2y} &\Rightarrow 2y = \frac{7}{x} \Rightarrow y = \frac{7}{2x} \\ f^{-1}(x) &= \frac{7}{2x}. \end{aligned}$$

So $f^{-1}(x) = \frac{7}{2x}$, $x \neq 0$.

2 d)

$$\begin{aligned} y &= \frac{2x+1}{x-3} \\ x &= \frac{2y+1}{y-3} \Rightarrow x(y-3) = 2y+1 \Rightarrow xy - 3x = 2y+1 \\ &\Rightarrow xy - 2y = 3x+1 \Rightarrow y(x-2) = 3x+1 \\ &\Rightarrow y = \frac{3x+1}{x-2} \\ f^{-1}(x) &= \frac{3x+1}{x-2}. \end{aligned}$$

So $f^{-1}(x) = \frac{3x+1}{x-2}$, $x \neq 2$.

2 e)

$$\begin{aligned} y &= \frac{4x-9}{2x+1} \\ x &= \frac{4y-9}{2y+1} \Rightarrow x(2y+1) = 4y-9 \Rightarrow 2xy + x = 4y - 9 \\ &\Rightarrow x + 9 = 4y - 2xy \Rightarrow x + 9 = y(4-2x) \\ &\Rightarrow y = \frac{x+9}{4-2x} \\ f^{-1}(x) &= \frac{x+9}{4-2x}. \end{aligned}$$

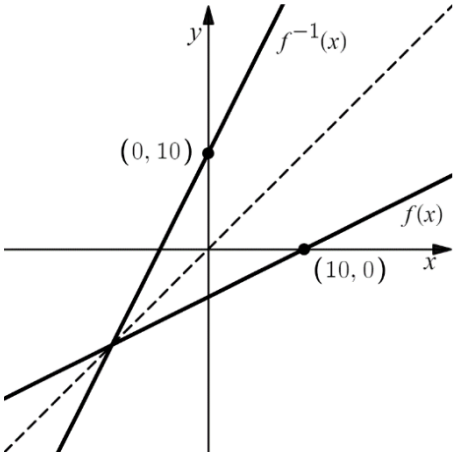
So $f^{-1}(x) = \frac{x+9}{4-2x}$, $x \neq 2$.

3 a)

$$\begin{aligned} y &= \frac{1}{2}x - 5 \\ x &= \frac{1}{2}y - 5 \Rightarrow x + 5 = \frac{1}{2}y \Rightarrow y = 2x + 10 \\ f^{-1}(x) &= 2x + 10. \end{aligned}$$

So $f^{-1}(x) = 2x + 10$, $x \in \mathbb{R}$.

3 b)



3 c)

$$f(12) = \frac{1}{2} \times 12 - 5 = 1$$
$$f^{-1}(1) = 2 \times 1 + 10 = 12.$$

The point $(12, 1)$ on f reflected across the line $y = x$ is the point $(1, 12)$ on f^{-1}.

4)

A reflection across the line $y = x$.

5 a)

$(3, 1)$.

5 b)

$(4, -5)$.

5 c)

$(-3e^2, 3)$.

5 d)

(b, a).

6 a)

$$x = 5x - 2 \Rightarrow 2 = 4x \Rightarrow x = \frac{1}{2}.$$

6 b)

$$x = -x + 4 \Rightarrow 2x = 4 \Rightarrow x = 2.$$

6 c)

$$x = x^2 - 6 \Rightarrow x^2 - x - 6 = 0 \Rightarrow x = -2,\ x = 3.$$

$x \geq 0$ so $x = 3$.

6 d)

$$\begin{aligned} x = -2x^2 - x + 4 &\Rightarrow 2x^2 + 2x - 4 = 0 \\ &\Rightarrow x = 1,\ x = -2. \end{aligned}$$

$x \geq 0$ so $x = 1$.

6 e)

$$\begin{aligned} x = x^3 &\Rightarrow x^3 - x = 0 \Rightarrow x(x^2 - 1) = 0 \\ &\Rightarrow x(x-1)(x+1) = 0 \Rightarrow x = 0,\ x = 1,\ x = -1. \end{aligned}$$

$x \geq 0$ so $x = 0$ or $x = 1$.

6 f)

$$\begin{aligned} x = \frac{2}{x+1} &\Rightarrow x(x+1) = 2 \Rightarrow x^2 + x = 2 \\ &\Rightarrow x^2 + x - 2 = 0 \Rightarrow x = -2,\ x = 1. \end{aligned}$$

$x \geq 0$ so $x = 1$.

7)

It is not one-to-one.

8 a)

$$\begin{aligned} ff(x) = f(-x + 12) &= -(-x + 12) + 12 \\ &= x - 12 + 12 = x \quad \text{as required.} \end{aligned}$$

8 b)

$$f^{-1}(x) = -x + 12, \quad x \neq 12.$$

9 a)

$$ff(x) = f(\frac{1}{x}) = \frac{1}{1/x} = \frac{x}{1} = x \quad \text{as required.}$$

9 b)

$$f^{-1}(x) = \frac{1}{x}, \quad x \neq 0.$$

10 a)

$$\begin{aligned} ff(x) = f(\frac{7x-48}{x-7}) &= \frac{7(\frac{7x-48}{x-7}) - 48}{(\frac{7x-48}{x-7}) - 7} \\ &= \frac{7(7x-48) - 48(x-7)}{(7x-48) - 7(x-7)} \\ &= \frac{49x - 336 - 48x + 336}{7x - 48 - 7x + 49} = \frac{x}{1} = x \quad \text{as required.} \end{aligned}$$

10 b)

$$f^{-1}(x) = \frac{7x-48}{x-7}, \quad x \neq 7.$$

11 a)

$$\begin{aligned} ff(x) = f(\frac{ax-1}{x-a}) &= \frac{a(\frac{ax-1}{x-a}) - 1}{(\frac{ax-1}{x-a}) - a} \\ &= \frac{a(ax-1) - 1(x-a)}{(ax-1) - a(x-a)} \\ &= \frac{a^2x - a - x + a}{ax - 1 - ax + a^2} = \frac{a^2x - x}{a^2 - 1} \\ &= \frac{x(a^2-1)}{a^2-1} = x \quad \text{as required.} \end{aligned}$$

11 b)

$$f^{-1}(x) = \frac{ax-1}{x-a}, \quad x \neq a.$$

12 a)

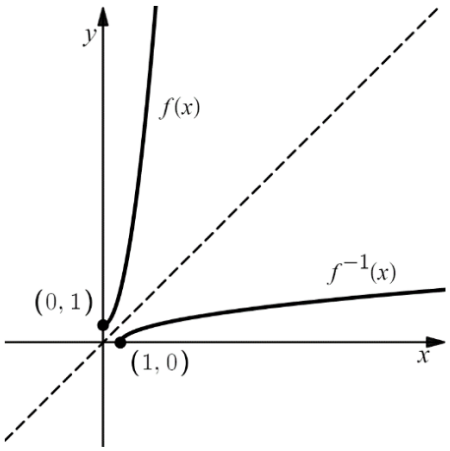
$$\begin{aligned} y &= 2x^2 + 1 \\ x &= 2y^2 + 1 \Rightarrow x - 1 = 2y^2 \Rightarrow y^2 = \frac{1}{2}x - \frac{1}{2} \\ x \geq 0 \text{ so } y &= \sqrt{\frac{1}{2}x - \frac{1}{2}} \\ f^{-1}(x) &= \sqrt{\frac{1}{2}x - \frac{1}{2}}, \quad x \geq 1. \end{aligned}$$

12 b)

$$\begin{aligned} 2x^2 + 1 = x \Rightarrow 2x^2 - x + 1 = 0 \\ b^2 - 4ac = 1 - 4 \times 2 \times 1 = -7 < 0. \end{aligned}$$

which has no real solutions.

12 c)



12 d)

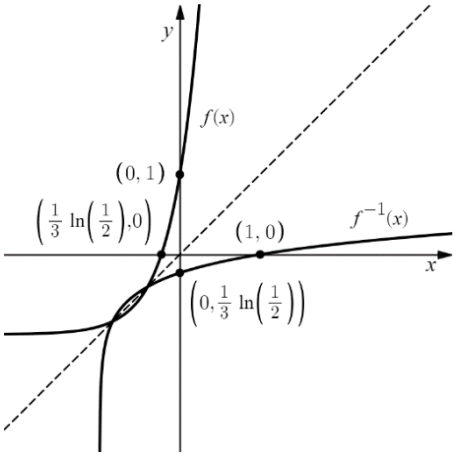
Domain: $\{x : x \in \mathbb{R}, x \geq 1\}$.

Range: $\{f^{-1}(x) : f^{-1}(x) \in \mathbb{R}, f^{-1}(x) \geq 0\}$.

13 a)

$$
\begin{aligned}
y &= 2e^{3x} - 1 \\
x = 2e^{3y} - 1 &\Rightarrow x + 1 = 2e^{3y} \Rightarrow e^{3y} = \frac{1}{2}x + \frac{1}{2} \\
&\Rightarrow 3y = \ln(\frac{1}{2}x + \frac{1}{2}) \Rightarrow y = \frac{1}{3}\ln(\frac{1}{2}x + \frac{1}{2}) \\
f^{-1}(x) &= \frac{1}{3}\ln(\frac{1}{2}x + \frac{1}{2}), \quad x > -1.
\end{aligned}
$$

13 b)



13 c)

Range of $f(x)$ is $f(x) > -1$, so

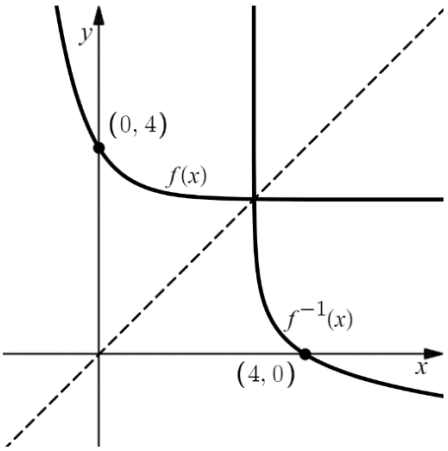
Domain: $\{x : x \in \mathbb{R}, x > -1\}$.

Range: $\{f^{-1}(x) : f^{-1}(x) \in \mathbb{R}\}$.

14 a)

$$\begin{aligned} y &= 3 + 5^{-x} \\ x &= 3 + 5^{-y} \Rightarrow x - 3 = 5^{-y} \Rightarrow \log_5(x-3) = -y \\ &\Rightarrow y = -\log_5(x-3) \\ f^{-1}(x) &= -\log_5(x-3), \quad x > 3. \end{aligned}$$

14 b)



14 c)

Range of $f(x)$ is $f(x) > 3$, so

Domain: $\{x : x \in \mathbb{R}, x > 3\}$.

Range: $\{f^{-1}(x) : f^{-1}(x) \in \mathbb{R}\}$.

15 a)

$$\begin{aligned} y &= \frac{2x}{x+5} \\ x &= \frac{2y}{y+5} \Rightarrow x(y+5) = 2y \Rightarrow xy + 5x = 2y \\ &\Rightarrow 5x = 2y - xy \Rightarrow 5x = y(2-x) \Rightarrow y = \frac{5x}{2-x} \\ f(1) &= \frac{1}{3} \quad \text{so} \\ f^{-1}(x) &= \frac{5x}{2-x}, \quad -8 \leq x \leq \frac{1}{3}. \end{aligned}$$

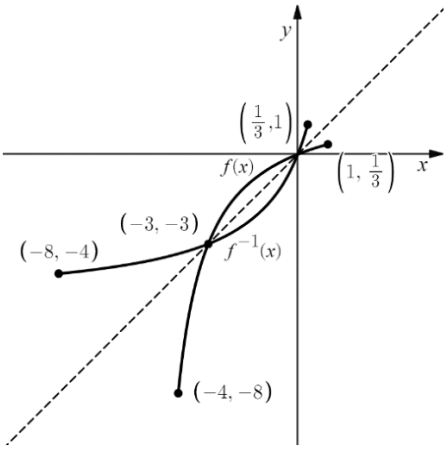
15 b)

$$\begin{aligned} \frac{2x}{x+5} = x &\Rightarrow 2x = x(x+5) \Rightarrow 2x = x^2 + 5x \\ &\Rightarrow x^2 + 3x = 0 \Rightarrow x(x+3) = 0 \\ x = 0 &\text{ or } x = -3. \end{aligned}$$

15 c)

$$f(-4) = -8, \quad f(1) = \frac{1}{3}.$$

$(-4, -8)$ to $(1, \frac{1}{3})$.



15 d)

Domain: $\{x : x \in \mathbb{R}, -8 \leq x \leq \frac{1}{3}\}$.

Range: $\{f^{-1}(x) : f^{-1}(x) \in \mathbb{R}, -4 \leq f^{-1}(x) \leq 1\}$.

16 a)

$$f(-1) = -7, \quad f(5) = 23.$$

Domain: $\{x : x \in \mathbb{R}, -7 \leq x \leq 23\}$.

Range: $\{f^{-1}(x) : f^{-1}(x) \in \mathbb{R}, -1 \leq f^{-1}(x) \leq 5\}$.

16 b)

Domain: $\{x : x \in \mathbb{R}\}$.

Range: $\{f^{-1}(x) : f^{-1}(x) \in \mathbb{R}, f^{-1}(x) > 0\}$.

16 c)

Domain: $\{x : x \in \mathbb{R}, x > 0\}$.

Range: $\{f^{-1}(x) : f^{-1}(x) \in \mathbb{R}\}$.

16 d)

Domain: $\{x : x \in \mathbb{R}, -1 \leq x \leq 1\}$.

Range: $\{f^{-1}(x) : f^{-1}(x) \in \mathbb{R}, -\frac{\pi}{2} \leq f^{-1}(x) \leq \frac{\pi}{2}\}$.

16 e)

Domain: $\{x : x \in \mathbb{R}, -1 \leq x \leq 1\}$.

Range: $\{f^{-1}(x) : f^{-1}(x) \in \mathbb{R}, 0 \leq f^{-1}(x) \leq \pi\}$.

16 f)

Domain: $\{x : x \in \mathbb{R}\}$.

Range: $\{f^{-1}(x) : f^{-1}(x) \in \mathbb{R}, -\frac{\pi}{2} < f^{-1}(x) < \frac{\pi}{2}\}$.

17 a)

$$f(-1) = (-1)^4 - (-1)^3 + 2(-1) = 1 + 1 - 2 = 0.$$

$\therefore (x + 1)$ is a factor.

17 b)

$$f(x) = x(x^3 - x^2 + 2).$$

	x^2	$-2x$	$+2$
x	x^3	$-2x^2$	$+2x$
$+1$	x^2	$-2x$	$+2$

$$f(x) = x(x^3 - x^2 + 2) = x(x + 1)(x^2 - 2x + 2).$$

$$(-2)^2 - 4 \times 1 \times 2 = -4 < 0.$$

So no solutions to $x^2 - 2x + 2 = 0$.

Only solutions to $f(x) = 0$ are $x = 0$ and $x = -1$.

17 c)

The function is not one-to-one.

17 d)

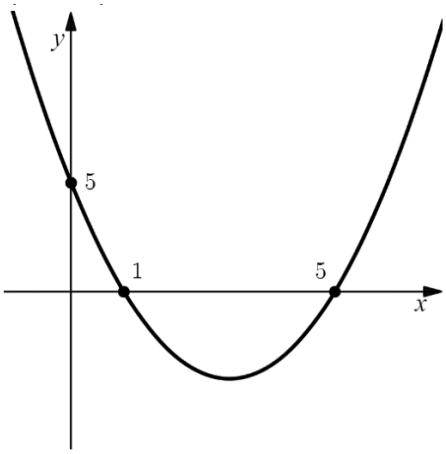
$$f(0) = 0 \text{ and } f(2) = 2^4 - 2^3 + 2(2) = 12.$$

Domain: $\{x : x \in \mathbb{R}, 0 \leq x \leq 12\}$.

Range: $\{f^{-1}(x) : f^{-1}(x) \in \mathbb{R}, 0 \leq f^{-1}(x) \leq 2\}$.

18 a)

$$f(x) = x^2 - 6x + 5 = (x-5)(x-1).$$



18 b)

Minimum point at $(3, -4)$ so $p = 3$.

18 c)

Range: $\{f(x) : f(x) \in \mathbb{R}, f(x) \geq -4\}$.

18 d)

Domain: $\{x : x \in \mathbb{R}, x \geq -4\}$.

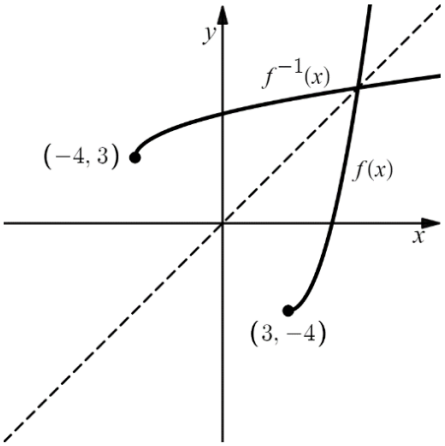
Range: $\{f^{-1}(x) : f^{-1}(x) \in \mathbb{R}, f^{-1}(x) \geq 3\}$.

18 e)

$$x^2 - 6x + 5 = x$$
$$\Rightarrow x^2 - 7x + 5 = 0 \Rightarrow x = \frac{7}{2} \pm \frac{1}{2}\sqrt{29}.$$

But $x \geq 3$ so $x = \frac{7}{2} + \frac{1}{2}\sqrt{29}$ only and therefore only one solution.

18 f)



19 a)

$$p = 3.$$

19 b)

$\{f(x) : f(x) \in \mathbb{R}, f(x) \in \mathbb{R}\}$.

19 c)

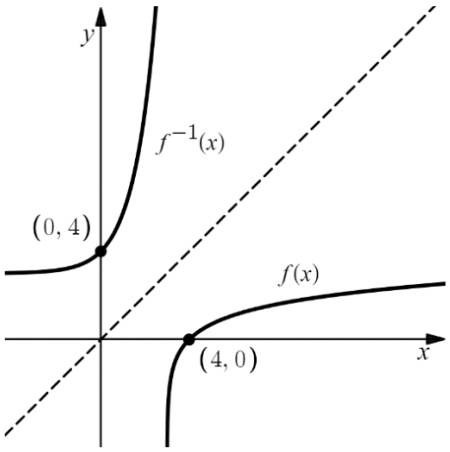
$$\begin{aligned} y &= \ln(x-3) \\ x &= \ln(y-3) \Rightarrow e^x = y-3 \Rightarrow e^x + 3 = y \\ &\Rightarrow y = e^x + 3 \\ f^{-1}(x) &= e^x + 3, \quad x \in \mathbb{R}. \end{aligned}$$

19 d)

Domain: $\{x : x \in \mathbb{R}, x \in \mathbb{R}\}$.

Range: $\{f^{-1}(x) : f^{-1}(x) \in \mathbb{R}, f^{-1}(x) > 3\}$.

19 e)



20 a)

$$p = 5.$$

20 b)

Range: $\{f(x) : f(x) \in \mathbb{R}, f(x) \geq 0\}$.

20 c)

$$\begin{aligned} y &= \sqrt{x-5} \\ x &= \sqrt{y-5} \Rightarrow x^2 = y - 5 \Rightarrow x^2 + 5 = y \\ f(5) &= 0 \quad \text{so} \\ f^{-1}(x) &= x^2 + 5, \quad x \geq 0. \end{aligned}$$

20 d)

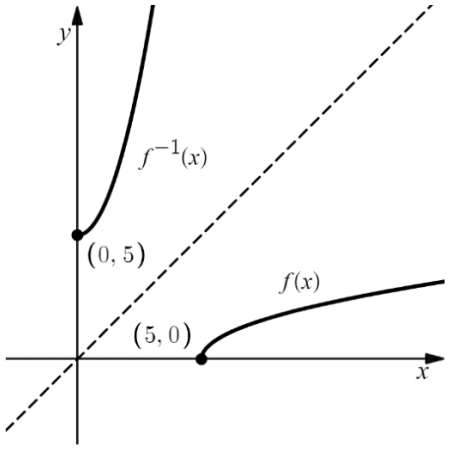
Domain: $\{x : x \in \mathbb{R}, x \geq 0\}$.

Range: $\{f^{-1}(x) : f^{-1}(x) \in \mathbb{R}, f^{-1}(x) \geq 5\}$.

20 e)

$$\begin{aligned} \sqrt{x-5} = x^2 + 5 &\Rightarrow x - 5 = (x^2 + 5)^2 \\ &\Rightarrow x - 5 = x^4 + 10x^2 + 25 \\ &\Rightarrow x^4 + 10x^2 - x + 30 = 0. \end{aligned}$$

20 f)



21 a)

$$p = -2.$$

21 b)

Range: $\{f(x) : f(x) \in \mathbb{R}, f(x) < 3\}$.

21 c)

$$\begin{aligned} y &= 3 - \frac{1}{x+2} \\ x &= 3 - \frac{1}{y+2} \Rightarrow \frac{1}{y+2} = 3 - x \Rightarrow \frac{1}{3-x} = y + 2 \\ &\Rightarrow y = \frac{1}{3-x} - 2 \\ f^{-1}(x) &= \frac{1}{3-x} - 2, \quad x \neq 3. \end{aligned}$$

21 d)

Domain: $\{x : x \in \mathbb{R}, x < 3\}$.

Range: $\{f^{-1}(x) : f^{-1}(x) \in \mathbb{R}, f^{-1}(x) > -2\}$.

21 e)

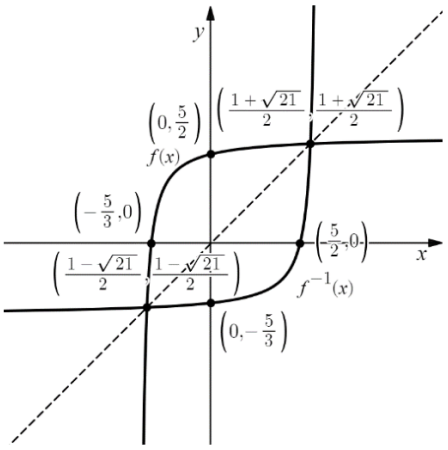
$$\begin{aligned} 3 - \frac{1}{x+2} = x &\Rightarrow 3(x+2) - 1 = x(x+2) \\ &\Rightarrow 3x + 6 - 1 = x^2 + 2x \Rightarrow x^2 - x - 5 = 0 \\ x &= \frac{1}{2} \pm \frac{1}{2}\sqrt{21}. \end{aligned}$$

21 f)

$f(0) = \frac{5}{2}$.

$f(x) = 0$ when

$$0 = 3 - \frac{1}{x+2} \Rightarrow \frac{1}{x+2} = 3 \Rightarrow \frac{1}{3} = x + 2 \Rightarrow x = -\frac{5}{3}.$$



22 a)

$$p = 5.$$

22 b)

$$\frac{4x-1}{x-5} = \frac{4(x-5)+19}{x-5} = 4 + \frac{19}{x-5}.$$

Range: $\{f(x) : f(x) \in \mathbb{R}, f(x) \neq 4\}$.

22 c)

Domain: $\{x : x \in \mathbb{R}, x \neq 4\}$.

Range: $\{f^{-1}(x) : f^{-1}(x) \in \mathbb{R}, f^{-1}(x) \neq 5\}$.

23 a)

$$1 = 5x + 3 \Rightarrow -2 = 5x \Rightarrow x = -\frac{2}{5}.$$

23 b)

$$1 = x^2 - 8 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3 \text{ so } x = 3.$$

23 c)

$$1 = \frac{4x+9}{2x+1} \Rightarrow 2x+1 = 4x+9 \Rightarrow -8 = 2x \Rightarrow x = -4.$$

24 a)

$$f\prime(x) = a.$$

24 b)

$$\begin{aligned} y &= ax + b \\ x &= ay + b \Rightarrow ay = x - b \Rightarrow y = \frac{1}{a}x - \frac{b}{a} \\ f^{-1}(x) &= \frac{1}{a}x - \frac{b}{a} \Rightarrow (f^{-1})\prime(x) = \frac{1}{a}. \end{aligned}$$

24 c)

$$a = \frac{1}{a} \Rightarrow a^2 = 1 \Rightarrow a = \pm 1.$$

25 a)

$$f\prime(x) = 2x \text{ so } f\prime(2) = 4.$$

25 b)

$$\begin{aligned} y &= x^2 \\ x &= y^2 \Rightarrow y = \sqrt{x} \\ f^{-1}(x) &= x^{1/2} \Rightarrow (f^{-1})\prime(x) = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}} \\ &\Rightarrow (f^{-1})\prime(2) = \frac{1}{2} \times \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{4}. \end{aligned}$$

25 c)

$$\begin{aligned} 2x = \frac{1}{2\sqrt{x}} &\Rightarrow 4x\sqrt{x} = 1 \Rightarrow x^{3/2} = \frac{1}{4} \\ &\Rightarrow x = (\frac{1}{4})^{2/3} = 4^{-2/3} = \frac{1}{4^{2/3}} = \frac{1}{2\sqrt[3]{2}}. \end{aligned}$$

26 a)

$$fg(x) = f(2x-1) = -(2x-1) + 5 = -2x + 6.$$

26 b)

$$\begin{aligned} y &= 2x - 1 \\ x &= 2y - 1 \Rightarrow x + 1 = 2y \Rightarrow y = \frac{1}{2}x + \frac{1}{2} \\ g^{-1}(x) &= \frac{1}{2}x + \frac{1}{2} \\ -2(5) + 6 &= -4 \\ g^{-1}(k) &= \frac{1}{2}k + \frac{1}{2} \\ \frac{1}{2}k + \frac{1}{2} &= -4 \Rightarrow k = -9. \end{aligned}$$

27 a)

$$\begin{aligned} fg(x) = f(5x+1) &= 2(5x+1)^2 - 8(5x+1) \\ &= 50x^2 + 20x + 2 - 40x - 8 = 50x^2 - 20x - 6. \end{aligned}$$

27 b)

$$\begin{aligned} y &= 5x + 1 \\ x &= 5y + 1 \Rightarrow x - 1 = 5y \Rightarrow y = \frac{1}{5}x - \frac{1}{5} \\ g^{-1}(x) &= \frac{1}{5}x - \frac{1}{5} \\ 50 \times 4 - 20 \times 2 - 6 &= 154 \\ g^{-1}(k) &= \frac{1}{5}k - \frac{1}{5} \\ \frac{1}{5}k - \frac{1}{5} &= 154 \Rightarrow k = 771. \end{aligned}$$

28 a)

$$\begin{aligned} gf(x) = g(-2x+1) &= 3(-2x+1) + 5 \\ &= -6x + 3 + 5 = -6x + 8. \end{aligned}$$

28 b)

$$-2x + 1 > 1 \Rightarrow -2x > 0 \Rightarrow x < 0.$$

Domain of $f(x)$ is $x < 0$.

Domain of gf: $\{x : x \in \mathbb{R}, x < 0\}$.

28 c)

$$y = -6x + 8.$$

Range of gf: $\{gf(x) : gf(x) \in \mathbb{R}, gf(x) > 8\}$.

28 d)

$$\begin{aligned} y &= -6x + 8 \\ x &= -6y + 8 \Rightarrow x - 8 = -6y \Rightarrow y = -\frac{1}{6}x + \frac{4}{3} \\ (gf)^{-1}(x) &= -\frac{1}{6}x + \frac{4}{3}, \quad x > 8. \end{aligned}$$

28 e)

Domain: $\{x : x \in \mathbb{R}, x > 8\}$.

Range: $\{(gf)^{-1}(x) : (gf)^{-1}(x) \in \mathbb{R}, (gf)^{-1}(x) < 0\}$.

29 a)

$$\begin{aligned} gf(x) &= g(x+4) = (x+4)^2 - (x+4) \\ &= x^2 + 8x + 16 - x - 4 = x^2 + 7x + 12. \end{aligned}$$

29 b)

$$x + 4 > 1 \Rightarrow x > -3.$$

Domain of $f(x)$ is $x < 0$.

Domain of gf: $\{x : x \in \mathbb{R}, -3 < x < 0\}$.

29 c)

$$gf(0) = 12.$$

Range of gf: $\{gf(x) : gf(x) \in \mathbb{R}, 0 < gf(x) < 12\}$.

29 d)

$$\begin{aligned} y &= x^2 + 7x + 12 \\ x &= y^2 + 7y + 12 \Rightarrow x - 12 = (y + 3.5)^2 - 3.5^2 \\ &\Rightarrow x + 0.25 = (y + 3.5)^2 \Rightarrow \sqrt{x + 0.25} = y + 3.5 \\ y &= -3.5 + \sqrt{x + 0.25} \\ (gf)^{-1}(x) &= -3.5 + \sqrt{x + 0.25}, \quad 0 < x < 12. \end{aligned}$$

29 e)

Domain: $\{x : x \in \mathbb{R}, 0 < x < 12\}$.

Range: $\{(gf)^{-1}(x) : (gf)^{-1}(x) \in \mathbb{R}, -3 < (gf)^{-1}(x) < 0\}$.

30 a)

$$\begin{aligned} fg(x) &= f(\frac{2x+2}{x+2}) = 3(\frac{2x+2}{x+2}) + 1 = \frac{6x+6}{x+2} + 1 \\ &= \frac{6(x+2) - 6}{x+2} + 1 = 7 - \frac{6}{x+2}. \end{aligned}$$

30 b)

For $g(x)$, $x \neq -2$.

Domain: $\{x : x \in \mathbb{R}, x \neq -2\}$.

30 c)

Range: $\{fg(x) : fg(x) \in \mathbb{R}, fg(x) \neq 7\}$.

30 d)

$$\begin{aligned} y &= 7 - \frac{6}{x+2} \\ x &= 7 - \frac{6}{y+2} \Rightarrow \frac{6}{y+2} = 7 - x \Rightarrow 6 = (7-x)(y+2) \\ &\Rightarrow 6 = 7y + 14 - xy - 2x \\ &\Rightarrow 2x - 8 = (7-x)y \Rightarrow y = \frac{2x-8}{7-x} \\ f^{-1}(x) &= \frac{2x-8}{7-x}, \quad x \neq 7. \end{aligned}$$

30 e)

Domain: $\{x : x \in \mathbb{R}, x \neq -7\}$.

Range: $\{(fg)^{-1}(x) : (fg)^{-1}(x) \in \mathbb{R}, (fg)^{-1}(x) \neq -2\}$.

31 a)

$$\begin{aligned} y &= x^2 - 6x + 1 = (x-3)^2 - 8 \\ \therefore f(1) &= 1 - 6 + 1 = -4. \end{aligned}$$

Min at $(3, -8)$ which is not within $x \leq 1$.

Range: $f(x) \in [-4, \infty)$.

31 b)

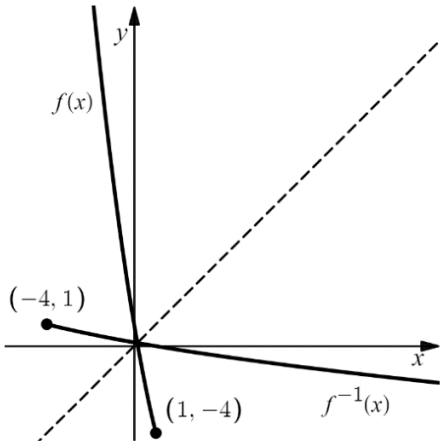
$$\begin{aligned} y &= x^2 - 6x + 1 \\ x &= y^2 - 6y + 1 \Rightarrow x = (y-3)^2 - 9 + 1 \\ &\Rightarrow x = (y-3)^2 - 8 \Rightarrow x + 8 = (y-3)^2 \\ &\Rightarrow -\sqrt{x+8} = y - 3 \Rightarrow y = 3 - \sqrt{x+8} \\ f^{-1}(x) &= 3 - \sqrt{x+8}, \quad x \in [-4, \infty). \end{aligned}$$

The domain of f is restricted to negative values of x.

31 c)

$$\begin{aligned} x &= x^2 - 6x + 1 \Rightarrow x^2 - 7x + 1 = 0 \\ &\Rightarrow x = \frac{7}{2} \pm \frac{3}{2}\sqrt{5} \text{ so as } x \leq 1, \quad x = \frac{7}{2} - \frac{3}{2}\sqrt{5}. \end{aligned}$$

31 d)



31 e)

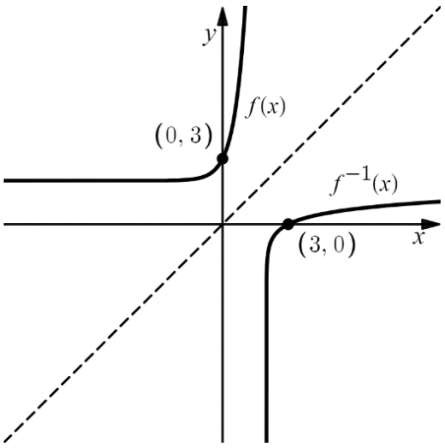
Domain: $x \in (-4, \infty)$.

Range: $f^{-1}(x) \in (-\infty, 1)$.

32 a)

$$\begin{aligned} y &= e^{2x} + 2 \\ x &= e^{2y} + 2 \Rightarrow x - 2 = e^{2y} \Rightarrow \ln(x-2) = 2y \\ &\Rightarrow y = \frac{1}{2}\ln(x-2) \\ f^{-1}(x) &= \frac{1}{2}\ln(x-2), \quad x \in (2, \infty). \end{aligned}$$

32 b)



32 c)

Domain: $x \in (2, \infty)$.

Range: $f^{-1}(x) \in \mathbb{R}$.

33 a)

$$y = \frac{3x}{x-4}$$

$$x = \frac{3y}{y-4} \Rightarrow x(y-4) = 3y \Rightarrow xy - 4x = 3y$$

$$\Rightarrow 4x = xy - 3y \Rightarrow 4x = y(x-3) \Rightarrow y = \frac{4x}{x-3}$$

$$f(-5) = \frac{5}{3}, \quad f(2) = -3$$

$$f^{-1}(x) = \frac{4x}{x-3}, \quad x \in [-3, \frac{5}{3}].$$

33 b)

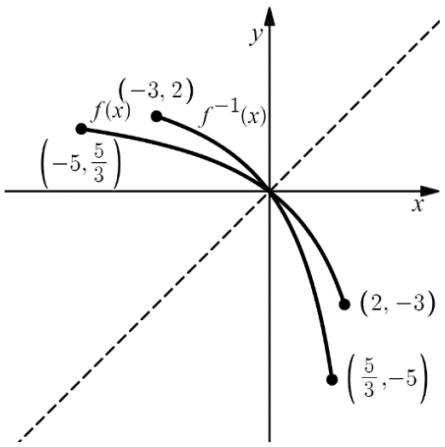
$$\frac{3x}{x-4} = x \Rightarrow 3x = x(x-4) \Rightarrow 3x = x^2 - 4x$$

$$\Rightarrow x^2 - 7x = 0 \Rightarrow x(x-7) = 0$$

$$x = 0 \text{ or } x = 7 \text{ but } -5 \leq x \leq 2 \text{ so } x = 0.$$

33 c)

$(-5, \frac{5}{3})$ to $(2, -3)$.



33 d)

Domain: $x \in [-3, \frac{5}{3}]$.

Range: $f^{-1}(x) \in [-5, 2]$.

34 a)

$$f(-2) = 7, \quad f(7) = -11.$$

Domain: $x \in [-11, 7]$.

Range: $f^{-1}(x) \in [-2, 7]$.

34 b)

Domain: $x \in (-\infty, \infty)$.

Range: $f^{-1}(x) \in (0, \infty)$.

34 c)

Domain: $x \in (2, \infty)$.

Range: $f^{-1}(x) \in (-\infty, \infty)$.

35 a)

$$p = -2.$$

35 b)

$$f(x) \in (-\infty, \infty).$$

35 c)

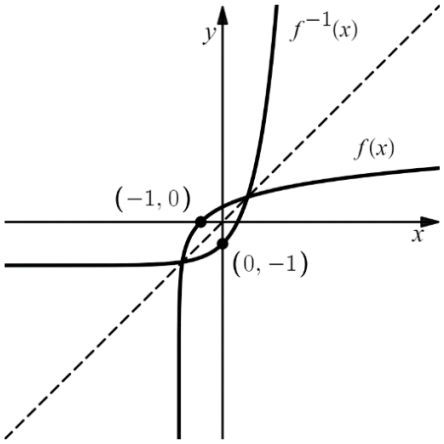
$$\begin{aligned} y &= \ln(x+2) \\ x &= \ln(y+2) \Rightarrow e^x = y+2 \Rightarrow e^x - 2 = y \\ f^{-1}(x) &= e^x - 2, \quad x \in \mathbb{R}. \end{aligned}$$

35 d)

Domain: $x \in (-\infty, \infty)$.

Range: $f^{-1}(x) \in (-2, \infty)$.

35 e)



36 a)

$$p = 9.$$

36 b)

$$\begin{aligned} \frac{5x+2}{x-9} = \frac{5(x-9)+47}{x-9} &= 5 + \frac{47}{x-9} \\ y &\neq 5 \\ f(x) &\in (-\infty, 5) \cup (5, \infty). \end{aligned}$$

36 c)

Domain: $x \in (-\infty, 5) \cup (5, \infty)$.

Range: $f^{-1}(x) \in (-\infty, 9) \cup (9, \infty)$.

37 a)

$$\begin{aligned} f(x) &= 2x^2 + 4x - 5 = 2(x^2 + 2x) - 5 \\ &= 2((x+1)^2 - 1) - 5 = 2(x+1)^2 - 7. \end{aligned}$$

Minimum point at $(-1, -7)$ so $p = -1$.

37 b)

Range of f is $f(x) \in [-7, \infty)$.

Domain of f^{-1}: $x \in [-7, \infty)$.

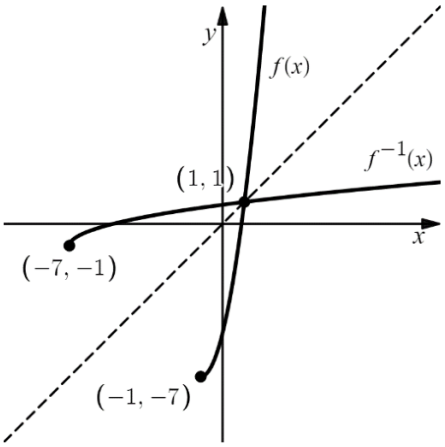
Range of f^{-1}: $f^{-1}(x) \in [-1, \infty)$.

37 c)

$$2x^2 + 4x - 5 = x$$

$$\Rightarrow 2x^2 + 3x - 5 = 0 \Rightarrow x = 1 \text{ or } x = -\frac{5}{2}.$$

But $x \geq -1$ so $x = 1$.



38 a)

$$gf(x) = g(1 + 2x) = (1 + 2x)^2 = 4x^2 + 4x + 1.$$

38 b)

$$1 + 2x > 2 \Rightarrow 2x > 1 \Rightarrow x > \frac{1}{2}.$$

Domain of $f(x)$ is $x < 1$.

Domain of gf: $x \in (\frac{1}{2}, 1)$.

38 c)

$$gf(0.5) = 4, \quad gf(1) = 9.$$

Range of gf: $gf(x) \in (4, 9)$.

38 d)

$$\begin{aligned}
y &= 4x^2 + 4x + 1 \\
x &= 4y^2 + 4y + 1 \Rightarrow x = 4(y^2 + y) + 1 \\
&\Rightarrow x = 4((y + \frac{1}{2})^2 - \frac{1}{4}) + 1 \\
&\Rightarrow x = 4(y + \frac{1}{2})^2 - 1 + 1 \\
&\Rightarrow \frac{1}{4}x = (y + \frac{1}{2})^2 \Rightarrow \frac{1}{2}\sqrt{x} = y + \frac{1}{2} \\
y &= -\frac{1}{2} + \frac{1}{2}\sqrt{x} \\
(gf)^{-1}(x) &= -\frac{1}{2} + \frac{1}{2}\sqrt{x}, \quad x \in (4, 9).
\end{aligned}$$

38 e)

Domain: $x \in (4, 9)$.

Range: $(gf)^{-1}(x) \in (\frac{1}{2}, 1)$.

39 a)

$$\begin{aligned}
fg(x) = f(\frac{8x + 21}{x + 7}) &= -2(\frac{8x + 21}{x + 7}) + 3 = \frac{-16x - 42}{x + 7} + 3 \\
&= \frac{-16(x + 7) + 70}{x + 7} + 3 = -13 + \frac{70}{x + 7}.
\end{aligned}$$

39 b)

For $g(x)$, $x \neq -7$.

Domain: $x \in (-\infty, -7) \cup (-7, \infty)$.

39 c)

Range: $fg(x) \in (-\infty, -13) \cup (-13, \infty)$.

39 d)

$$\begin{aligned}
y &= -13 + \frac{70}{x + 7} \\
x &= -13 + \frac{70}{y + 7} \Rightarrow \frac{70}{y + 7} = x + 13 \\
&\Rightarrow 70 = (x + 13)(y + 7) \\
&\Rightarrow 70 = xy + 7x + 13y + 91 \\
&\Rightarrow -21 - 7x = (x + 13)y \\
&\Rightarrow y = \frac{-21 - 7x}{x + 13} \\
&\Rightarrow (fg)^{-1}(x) = \frac{-21 - 7x}{x + 13}, \quad x \neq -13.
\end{aligned}$$

39 e)

Domain: $x \in (-\infty, -13) \cup (-13, \infty)$.

Range: $(fg)^{-1}(x) \in (-\infty, -7) \cup (-7, \infty)$.